

Lecture 12: Midterm Review

Instructor: Zijie Zhou, IEDA, HKUST

October 14, 2025

Review: Convex Optimization

- Showing a set is convex.
- Showing a function is convex.
- Lagrangian Duality.
- Weak/Strong Duality.
- KKT Condition.
- Unconstrained Optimization and Descent Algorithms.

Showing a set C is convex

- 1 *By Definition:* For any $x_1, x_2 \in C$, $0 \leq \theta \leq 1$, we have

$$\theta x_1 + (1 - \theta)x_2 \in C.$$

- 2 *Use convex functions (will cover later)*
- 3 *Show that C is obtained from simple convex sets (hyperplanes, halfspaces, norm balls, ...) by operations that preserve convexity*
- intersection
 - affine mapping
 - perspective mapping
 - linear-fractional mapping

Showing a set C is convex

Problem (Convexity by first principles)

Let

$$S = \left\{ x = (x_1, x_2) \in \mathbb{R}^2 \mid \|x\|_2 \leq 2 \text{ and } x_1 + x_2 \geq 0 \right\}.$$

- 1 *Decide whether S is convex.*
- 2 *Prove your answer directly from the definition of convexity: for arbitrary $x, y \in S$ and any $\theta \in [0, 1]$, show that $\theta x + (1 - \theta)y \in S$.*

Showing a set C is convex

Solution

Proof (by definition). Let $x, y \in S$ and $\theta \in [0, 1]$. We must show that

$z := \theta x + (1 - \theta)y \in S$, i.e., $\|z\|_2 \leq 2$ and $z_1 + z_2 \geq 0$.

(i) Euclidean ball constraint. By the triangle inequality and positive homogeneity of the Euclidean norm,

$$\|z\|_2 = \|\theta x + (1 - \theta)y\|_2 \leq \theta\|x\|_2 + (1 - \theta)\|y\|_2.$$

Since $x, y \in S$, we have $\|x\|_2 \leq 2$ and $\|y\|_2 \leq 2$, hence

$$\|z\|_2 \leq \theta \cdot 2 + (1 - \theta) \cdot 2 = 2.$$

(ii) Halfspace constraint. Linearity of addition gives

$z_1 + z_2 = \theta(x_1 + x_2) + (1 - \theta)(y_1 + y_2)$. Because $x, y \in S$, we have $x_1 + x_2 \geq 0$ and $y_1 + y_2 \geq 0$. With $\theta \in [0, 1]$, it follows that

$$z_1 + z_2 \geq \theta \cdot 0 + (1 - \theta) \cdot 0 = 0.$$

Thus z satisfies the halfspace constraint.

Combining (i) and (ii), we conclude $z \in S$. Since $x, y \in S$ and $\theta \in [0, 1]$ were arbitrary, S is convex by the definition of convexity. □

Convex Function Definition

- ▶ $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is convex if $\mathbf{dom} f$ is a convex set and for all $x, y \in \mathbf{dom} f$, $0 \leq \theta \leq 1$,

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$



- ▶ f is concave if $-f$ is convex
- ▶ f is strictly convex if $\mathbf{dom} f$ is convex and for $x, y \in \mathbf{dom} f$, $x \neq y$, $0 < \theta < 1$,

$$f(\theta x + (1 - \theta)y) < \theta f(x) + (1 - \theta)f(y)$$

Characterization of convex functions: Restriction to a line

- ▶ $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is convex if and only if the function $g : \mathbf{R} \rightarrow \mathbf{R}$,

$$g(t) = f(x + tv), \quad \mathbf{dom} \, g = \{t \mid x + tv \in \mathbf{dom} f\}$$

is convex (in t) for any $x \in \mathbf{dom} f$, $v \in \mathbf{R}^n$

- ▶ can check convexity of f by checking convexity of functions of one variable

First-order condition

- ▶ f is **differentiable** if $\text{dom } f$ is open and the gradient

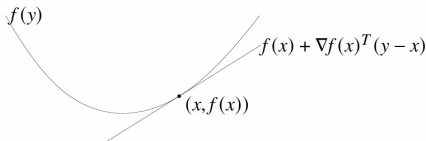
$$\nabla f(x) = \left(\frac{\partial f(x)}{\partial x_1}, \frac{\partial f(x)}{\partial x_2}, \dots, \frac{\partial f(x)}{\partial x_n} \right) \in \mathbf{R}^n$$

exists at each $x \in \text{dom } f$

- ▶ **1st-order condition:** differentiable f with convex domain is convex if and only if

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) \quad \text{for all } x, y \in \text{dom } f$$

- ▶ first order Taylor approximation of convex f is a **global underestimator** of f



Second-order conditions

- ▶ f is **twice differentiable** if $\text{dom} f$ is open and the Hessian $\nabla^2 f(x) \in \mathbf{S}^n$,

$$\nabla^2 f(x)_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}, \quad i, j = 1, \dots, n,$$

exists at each $x \in \text{dom} f$

- ▶ **2nd-order conditions:** for twice differentiable f with convex domain
 - f is convex if and only if $\nabla^2 f(x) \succeq 0$ for all $x \in \text{dom} f$
 - if $\nabla^2 f(x) \succ 0$ for all $x \in \text{dom} f$, then f is strictly convex

Second-order conditions

To check whether a Hessian matrix H is positive-semidefinite:

- If all eigenvalues of H are positive, then the function is convex.
- If all eigenvalues of H are negative, then the function is concave.
- If some eigenvalues are positive while others are negative, this function is neither convex nor concave.

Second-order conditions

Example (Finding eigenvalues of a 2×2 matrix)

Consider the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}.$$

We wish to find all eigenvalues λ satisfying $Av = \lambda v$, for some nonzero vector $v \in \mathbb{R}^2$.

Step 1. Characteristic equation. The eigenvalues satisfy

$$\det(A - \lambda I) = 0.$$

Compute

$$\det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 3 - \lambda \end{pmatrix} = (2 - \lambda)(3 - \lambda) - 1.$$

Step 2. Simplify and solve for λ .

$$(2 - \lambda)(3 - \lambda) - 1 = \lambda^2 - 5\lambda + 5 = 0.$$

Solve the quadratic equation: $\lambda = \frac{5 \pm \sqrt{25 - 20}}{2} = \frac{5 \pm 1}{2}$. Hence the eigenvalues are

$$\lambda_1 = 2, \quad \lambda_2 = 3.$$

Problem (A simple linear optimization and its dual)

Consider the primal linear program

$$\begin{aligned} (P) \quad & \min_{x \in \mathbb{R}^2} && 3x_1 + x_2 \\ & \text{s.t.} && x_1 + x_2 \geq 2, \\ & && x_1 + 2x_2 \geq 3, \\ & && x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

- (i) *Derive the Lagrangian dual problem.*
- (ii) *Solve both primal and dual, and check whether strong duality holds.*
- (iii) *State the KKT conditions and verify them at the optimal solutions.*

Lagrangian duality

- ▶ **standard form problem** (not necessarily convex)

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

variable $x \in \mathbf{R}^n$, domain $\mathcal{D}$, optimal value $p^\star$

- ▶ **Lagrangian:** $L : \mathbf{R}^n \times \mathbf{R}^m \times \mathbf{R}^p \rightarrow \mathbf{R}$, with $\text{dom } L = \mathcal{D} \times \mathbf{R}^m \times \mathbf{R}^p$,

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \nu_i h_i(x)$$

- weighted sum of objective and constraint functions
- λ_i is **Lagrange multiplier** associated with $f_i(x) \leq 0$
- ν_i is Lagrange multiplier associated with $h_i(x) = 0$

Lagrangian duality

(i) Lagrangian dual. Write the primal as

$$\min_{x \in \mathbb{R}^2} c^\top x \quad \text{s.t.} \quad Ax \geq b, \quad -x \leq 0,$$

with $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, $b = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, $c = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$. Introduce multipliers $y \in \mathbb{R}_{\geq 0}^2$ for $Ax \geq b$ and $s \in \mathbb{R}_{\geq 0}^2$ for $-x \leq 0$. The Lagrangian is

$$\mathcal{L}(x, y, s) = c^\top x + y^\top (b - Ax) + s^\top (-x) = b^\top y + (c - A^\top y - s)^\top x.$$

$$g(y, s) = \inf_{x \in \mathbb{R}^2} \mathcal{L}(x, y, s) = \begin{cases} b^\top y, & \text{if } c - A^\top y - s = 0, \\ -\infty, & \text{otherwise.} \end{cases}$$

Hence the dual is

$$\begin{aligned} \max_{y, s} \quad & b^\top y \\ \text{s.t.} \quad & c - A^\top y - s = 0, \\ & y \geq 0, \quad s \geq 0. \end{aligned}$$

$$\begin{aligned} \max_{y \in \mathbb{R}^2} \quad & 2y_1 + 3y_2 \\ \text{s.t.} \quad & A^\top y \leq c \iff \begin{cases} y_1 + y_2 \leq 3, \\ y_1 + 2y_2 \leq 1, \end{cases} \\ & y_1 \geq 0, \quad y_2 \geq 0. \end{aligned}$$

Strong Duality

(ii) Solve (P) and (D), check strong duality.

Primal. Because the objective is increasing in both x_1 and x_2 , an optimal solution lies on the boundary of the feasible region. Setting $x_1 = 0$ minimizes the high-cost variable; the constraints then require $x_2 \geq \max\{2, \frac{3}{2}\} = 2$. Thus $x^* = (0, 2)$ is feasible with value $3 \cdot 0 + 1 \cdot 2 = 2$. One can also check the intersection point $(1, 1)$ yields value $4 > 2$, hence

$$p^* = \min(P) = 2 \quad \text{attained at } x^* = (0, 2).$$

Dual. The feasible set is given by $y \geq 0$, $y_1 + y_2 \leq 3$, $y_1 + 2y_2 \leq 1$. Putting $y_2 = 0$ makes the second constraint $y_1 \leq 1$; take $y^* = (1, 0)$ which is feasible and gives objective $2 \cdot 1 + 3 \cdot 0 = 2$. No feasible point can exceed this because $y_1 + 2y_2 \leq 1$ implies $2y_1 + 3y_2 \leq 2$ by multiplying the second constraint by 1 and noting $y_2 \geq 0$. Hence

$$d^* = \max(D) = 2 \quad \text{attained at } y^* = (1, 0).$$

Strong duality. This is a linear program with finite primal and dual optimal values and nonempty feasible regions; we found $p^* = d^* = 2$. Therefore strong duality holds.

KKT condition

- KKT conditions:

$$f_i(\mathbf{x}^*) \leq 0, \quad i = 1, \dots, m \quad (\text{primal feasibility})$$

$$h_i(\mathbf{x}^*) = 0, \quad i = 1, \dots, p$$

$$\lambda_i^* \geq 0, \quad i = 1, \dots, m \quad (\text{dual feasibility})$$

$$\lambda_i^* f_i(\mathbf{x}^*) = 0, \quad i = 1, \dots, m \quad (\text{complementary slackness})$$

$$\nabla f_0(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(\mathbf{x}^*) + \sum_{i=1}^p \nu_i^* \nabla h_i(\mathbf{x}^*) = \mathbf{0} \quad (\text{zero Lagrangian gradient})$$

Theorem: KKT optimality conditions

For any optimization problem with differentiable functions:

- For any optimization problem (not necessarily convex) with strong duality, the KKT conditions are necessary for optimality.
- For a convex optimization problem satisfying Slater's condition, strong duality follows, and the KKT conditions are necessary and sufficient for optimality.

KKT Conditions

(iii) KKT Conditions. The KKT conditions are:

Primal feasibility:

$$Ax^* \geq b, \quad x^* \geq 0.$$

Dual feasibility:

$$A^T y^* \leq c, \quad y^* \geq 0.$$

Complementary slackness:

$$y_i^* (b_i - (Ax^*)_i) = 0, \quad i = 1, 2,$$

$$x_j^* (c_j - (A^T y^*)_j) = 0, \quad j = 1, 2.$$

Stationarity (zero Lagrangian gradient): $\nabla_x \mathcal{L}(x^*, y^*, s^*) = c - A^T y^* - s^* = 0.$

Verify at $x^* = (0, 2)$, $y^* = (1, 0)$:

Primal feasibility: $Ax^* = (2, 4)^T \geq (2, 3)^T = b$, and $x^* \geq 0$.

Dual feasibility: $A^T y^* = (1, 1)^T \leq (3, 1)^T = c$, and $y^* \geq 0$.

Complementary slackness (constraints):

$$b - (Ax^*) = (0, -1)^T \Rightarrow y_1^* \cdot 0 = 0, \quad y_2^* \cdot (-1) = 0.$$

Complementary slackness (variables):

$$c - A^T y^* = (2, 0)^T \Rightarrow x_1^* \cdot 2 = 0, \quad x_2^* \cdot 0 = 0.$$

Zero Lagrangian Gradient: $c - A^T y^* - s^* = 0$ by construction.

Descent method

- ▶ **descent methods** generate iterates as

$$x^{(k+1)} = x^{(k)} + t^{(k)} \Delta x^{(k)}$$

with $f(x^{(k+1)}) < f(x^{(k)})$ (hence the name)

- ▶ other notations: $x^+ = x + t\Delta x$, $x := x + t\Delta x$
- ▶ $\Delta x^{(k)}$ is the **step**, or **search direction**
- ▶ $t^{(k)} > 0$ is the **step size**, or **step length**
- ▶ from convexity, $f(x^+) < f(x)$ implies $\nabla f(x)^T \Delta x < 0$
- ▶ this means Δx is a **descent direction**

Descent method

General descent method.

given a starting point $x \in \text{dom} f$.

repeat

1. Determine a descent direction Δx .
2. **Line search.** Choose a step size $t > 0$.
3. **Update.** $x := x + t\Delta x$.

until stopping criterion is satisfied.

Gradient descent method

- ▶ general descent method with $\Delta x = -\nabla f(x)$

given a starting point $x \in \text{dom} f$.

repeat

1. $\Delta x := -\nabla f(x)$.
2. **Line search.** Choose step size t via exact or backtracking line search.
3. **Update.** $x := x + t\Delta x$.

until stopping criterion is satisfied.

- ▶ stopping criterion usually of the form $\|\nabla f(x)\|_2 \leq \epsilon$
- ▶ convergence result: for strongly convex f ,

$$f(x^{(k)}) - p^\star \leq c^k (f(x^{(0)}) - p^\star)$$

$c \in (0, 1)$ depends on m , $x^{(0)}$, line search type

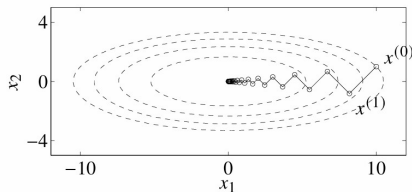
- ▶ very simple, but can be very slow

Gradient descent method example

- ▶ take $f(x) = (1/2)(x_1^2 + \gamma x_2^2)$, with $\gamma > 0$
- ▶ with exact line search, starting at $x^{(0)} = (\gamma, 1)$:

$$x_1^{(k)} = \gamma \left(\frac{\gamma - 1}{\gamma + 1} \right)^k, \quad x_2^{(k)} = \left(-\frac{\gamma - 1}{\gamma + 1} \right)^k$$

- very slow if $\gamma \gg 1$ or $\gamma \ll 1$
- example for $\gamma = 10$ at right
- called **zig-zagging**



Review: Portfolio Optimization

- Portfolio State Evolution.
- Portfolio Constraints.
- Performance Measures.
- Heuristics.
- Robust Portfolio Formulations.

Review: Portfolio Optimization

Consider three risky assets D, E, F with expected returns and covariance matrix

$$\mu = \begin{bmatrix} 0.18 \\ 0.12 \\ 0.22 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 0.60 & 0.15 & -0.05 \\ 0.15 & 0.50 & 0.10 \\ -0.05 & 0.10 & 0.90 \end{bmatrix}.$$

The risk-free rate is $r_f = 4\%$. Short selling is not allowed (weights are nonnegative and sum to 1).

(i) Compute the **expected return**, **volatility** and **Sharpe ratio** for the following portfolio:

$$Q_1 : (w_D, w_E, w_F) = (0.2, 0.5, 0.3).$$

Solution

For any weight vector w , the expected return is $ER = w^\top \mu$, the variance is $w^\top \Sigma w$, the volatility is $V = \sqrt{w^\top \Sigma w}$, and the Sharpe ratio is $SR = (ER - r_f)/V$.

Portfolio Q_1 : $w = (0.2, 0.5, 0.3)$

$$ER_1 = 0.2(0.18) + 0.5(0.12) + 0.3(0.22) = 0.162.$$

$$w^\top \Sigma w = 0.284 \quad \Rightarrow \quad V_1 = \sqrt{0.284} \approx 0.5329.$$

$$SR_1 = \frac{0.162 - 0.04}{0.5329} \approx 0.2289.$$

Review: Portfolio Optimization

Heuristic allocation rule: Select the top two assets by *mean variance score*

$$\text{score}_i = \frac{\mu_i}{\Sigma_{ii}},$$

then invest equally in those two assets. Which two assets are selected? What is the resulting portfolio and its **expected return**?

Review: Portfolio Optimization

Solution

Compute scores using the diagonal of Σ :

$$\text{score}_D = \frac{0.18}{0.60} = 0.300, \quad \text{score}_E = \frac{0.12}{0.50} = 0.240,$$

$$\text{score}_F = \frac{0.22}{0.90} \approx 0.2444.$$

Ranking: $D (0.300) > F (0.2444) > E (0.240)$. Top two are D and F . Thus the heuristic portfolio is

$$w_D = 0.5, \quad w_E = 0, \quad w_F = 0.5.$$

Its expected return is

$$ER = 0.5 \cdot 0.18 + 0.5 \cdot 0.22 = 0.20.$$

Review: Robust Optimization

- Robust Counterparts for Box, Ellipsoid, Polyhedron, Budget.
- Robust Counterparts for L_p Norm.
- Convex Conjugate of Functions.
- Robust Counterparts for Intersection of Convex Regions.

Convex Conjugate Review

$$f^*(y) = \sup_{x \in \text{dom}(f)} \{y^T x - f(x)\}.$$

Example: $f(x) = a^T x - b$, $x \in \mathbb{R}^n$.

We have:

$$f^*(y) = \sup_x \{y^T x - a^T x + b\} = \sup_x \{(y - a)^T x + b\}.$$

So:

$$f^*(y) = \begin{cases} b & y = a \\ \infty & y \neq a. \end{cases}$$

Robust Counterpart Table

$$(\bar{a} + Pz)^T x \leq b, \quad \forall z \in Z.$$

Unc. set	Z	Robust Counterpart	Tractability
Box	$\ z\ _\infty \leq \rho$	$\bar{a}^T x + \rho \ P^T x\ _1 \leq b$	LO
Ellipsoidal	$\ z\ _2 \leq \rho$	$\bar{a}^T x + \rho \ P^T x\ _2 \leq b$	CQO
Polyhedral	$Dz \leq d$	$\begin{cases} \bar{a}^T x + d^T y \leq b \\ D^T y = P^T x \\ y \geq 0 \end{cases}$	LO
Budget	$\begin{cases} \ z\ _\infty \leq \rho \\ \ z\ _1 \leq \Gamma \end{cases}$	$\bar{a}^T x + \rho \ y\ _1 + \Gamma \ P^T x - y\ _\infty \leq b$	LO
Convex constr.	$\begin{aligned} h_k(z) &\leq 0 \\ k &= 1, \dots, K \end{aligned}$	$\begin{cases} \bar{a}^T x + \sum_k u_k h_k^* \left(\frac{w^k}{u_k} \right) \leq b \\ \sum_k w^k = P^T x \\ u \geq 0 \end{cases}$	Convex

Computing δ^* : p -norm

Let $Z = \{z \mid \|z\|_p \leq \rho\}$, where $\|\cdot\|_p$ is the p -norm.

For the support function of Z we have:

$$\delta^*(y|Z) = \sup_{x : \|x\|_p \leq \rho} y^T x = \rho \|y\|_p^* = \rho \|y\|_q,$$

where $\|\cdot\|_p^*$ is the dual norm of the p -norm, i.e., $1/p + 1/q = 1$.

p	q
2	2
1	∞
∞	1

Hence, $\bar{a}^T x + \delta^*(P^T x|Z) \leq b$ is equivalent to: $\bar{a}^T x + \rho \|P^T x\|_q \leq b$.

Computing δ^* : intersection

$$(RC) \quad \bar{a}^T x + \delta^*(P^T x | Z) \leq b.$$

Lemma

Let $Z_1, \dots, Z_k$ be closed convex sets, such that $\bigcap_i \text{ri}(Z_i) \neq \emptyset$, and let $Z = \bigcap_{i=1}^k Z_i$. Then

$$\delta^*(y|Z) = \min_{y^1, \dots, y^k} \left\{ \sum_{i=1}^k \delta^*(y^i | Z_i) \mid \sum_{i=1}^k y^i = y \right\}.$$

and (RC) becomes

$$\begin{cases} \bar{a}^T x + \sum_{i=1}^k \delta^*(y^i | Z_i) \leq 0 \\ \sum_{i=1}^k y^i = P^T x \end{cases}$$

Proof: Use $\delta^*(y|Z) = (\delta(y|Z_1) + \dots + \delta(y|Z_k))^*$.

Computing δ^* : Ball-Box

Let $Z_2 = \{z \mid \|z\|_2 \leq \rho_2\}$ and $Z_\infty = \{z \mid \|z\|_\infty \leq \rho_\infty\}$.

Let $Z = Z_2 \cap Z_\infty$.

Then

$$\begin{aligned}\delta^*(y|Z) &= \min_{w^2, w^\infty} \{\delta^*(w^2|Z_2) + \delta^*(w^\infty|Z_\infty) \mid w^2 + w^\infty = y\} \\ &= \min_{w^2, w^\infty} \{\rho_2 \|w^2\|_2 + \rho_\infty \|w^\infty\|_1 \mid w^2 + w^\infty = y\}.\end{aligned}$$

Hence, $\bar{a}^T x + \delta^*(P^T x|Z) \leq b$ is equivalent to

$$\begin{cases} \bar{a}^T x + \rho_2 \|w^2\|_2 + \rho_\infty \|w^\infty\|_1 \leq b \\ w^2 + w^\infty = P^T x. \end{cases}$$